ECE401 MID RC

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Overview

Transformation Random Variables

Reliability

11.1. Theorem. Let $(\boldsymbol{X}, f_{\boldsymbol{X}})$ be a continuous multivariate random variable and let $\varphi\colon \mathbb{R}^n \to \mathbb{R}^n$ be a differentiable, bijective map with inverse φ^{-1}. Then $\boldsymbol{Y} = \varphi \circ \boldsymbol{X}$ is a continuous multivariate random variable with density

$$f_{\boldsymbol{Y}}(y) = f_{\boldsymbol{X}} \circ \varphi^{-1}(y) \cdot |\det D\varphi^{-1}(y)|,$$

where $D\varphi^{-1}$ is the Jacobian of φ^{-1}.

11.2. Lemma. Let $((X, Y), f_{XY})$ be a continuous bivariate random variable. Let $U = X/Y$. Then the density f_U of U is given by

$$f_U(u) = \int_{-\infty}^{\infty} f_{XY}(uv, v) \cdot |v|\, dv.$$

Proof.

Suppose that X and Y are independent and have probability density functions g and h respectively.

1. If X and Y have discrete distributions then $Z = X + Y$ has a discrete distribution with probability density function $g * h$ given by

$$(g * h)(z) = \sum_{x \in D_z} g(x)h(z - x), \quad z \in T$$

2. If X and Y have continuous distributions then $Z = X + Y$ has a continuous distribution with probability density function $g * h$ given by

$$(g * h)(z) = \int_{D_z} g(x)h(z - x)\, dx, \quad z \in T$$

In both cases, the probability density function $g * h$ is called the *convolution* of g and h.
Suppose that (X, Y) has a continuous distribution on $\mathbb{R}^2$ with probability density function f.

1. Random variable $V = XY$ has probability density function

$$v \mapsto \int_{-\infty}^{\infty} f(x, v/x) \frac{1}{|x|} dx$$

2. Random variable $W = Y/X$ has probability density function

$$w \mapsto \int_{-\infty}^{\infty} f(x, wx)|x| dx$$

Let (X, Y) be a continuous bivariate random variable with density $f_{XY} \colon S \to \mathbb{R}^2$ given by

$$f_{XY}(x, y) = \begin{cases} c \cdot (x^2 + y^2) & \text{for } x^2 + y^2 \leq 1, \\ 0 & \text{otherwise,} \end{cases}$$

where $c \in \mathbb{R}$ is a suitable constant.

i) Determine the constant c.

ii) Find $\mathrm{E}[X]$ and $\mathrm{E}[Y]$.

iii) Find $\mathrm{Var}[X]$ and $\mathrm{Var}[Y]$.

iv) Find the correlation coefficient ϱ_{XY}.

v) Find the density of $U = X/Y$.

Helpful note: $\int_0^{2\pi} \cos^2(\theta)\, d\theta = \pi$.

Exercise 1 (10 pts) Let $X_1, X_2 \sim \text{Exponential}(\lambda)$ be iid rv's, with density

$$f_X(x) = \lambda e^{-\lambda x}, \qquad x > 0, \qquad \lambda > 0$$

Find the probability density function of $Y = \max(X_1, X_2)$.

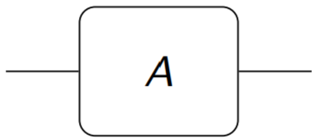
Exercise 5 (20 pts) Given function $g : \mathbb{R} \to \mathbb{R}$ as

$$g(x) = \begin{cases} 1, & -2 \le x < -1 \\ x^2, & -1 \le x < 0 \\ x, & 0 \le x < 2 \\ 2, & 2 \le x < 3 \\ 0, & \text{otherwise} \end{cases}$$

If $Y = g(X)$, and $X \sim \text{Uniform}([-4, 4])$ (i.e., uniform distribution over $[-4, 4]$), find the CDF of Y.

A Black Box System

Consider a "black box" unit A:



We don't care what the unit A does or what it looks like inside. We simply assume that at time $t = 0$ the unit A is working. Then at any time $t > 0$, either

- A is working or
- A has failed.

When A fails, it fails completely and can not be repaired.

The time when A fails is random; we describe it by the continuous random variable T_A. The density of T_A is called the

failure density f_A.

We denote by F_A the cumulative distribution function of T_A.

Failure Density

$$
\begin{aligned}
f_A(t) &= F_A'(t) \\
&= \lim_{\Delta t \to 0} \frac{F_A(t+\Delta t) - F_A(t)}{\Delta t} \\
&= \lim_{\Delta t \to 0} \frac{P[t \le T \le t + \Delta t]}{\Delta t}
\end{aligned}
$$

Reliability Function

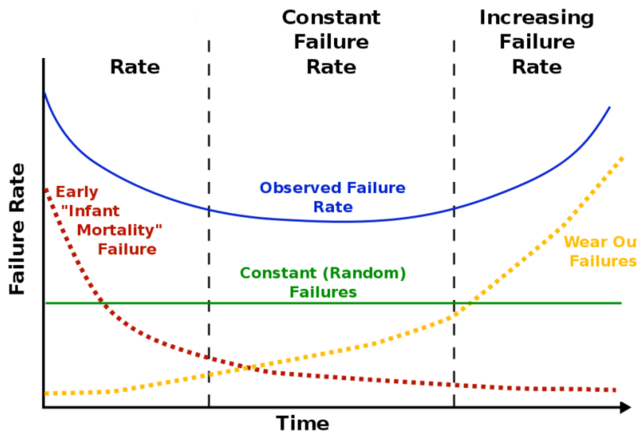
$$
\begin{aligned}
R_A(t) &= 1 - P[\text{component } A \text{ fails before time } t] \\
&= 1 - \int_0^t f_A(s)\, ds \\
&= 1 - F_A(t).
\end{aligned}
$$

Hazard Rate

$$\varrho_A(t) := \lim_{\Delta t \to 0} \frac{P[t \le T \le t + \Delta t \mid t \le T]}{\Delta t}$$

$$
\begin{aligned}
\varrho_A(t) &= \lim_{\Delta t \to 0} \frac{P[t \le T \le t + \Delta t]}{P[T \ge t] \cdot \Delta t} \\
&= \frac{f_A(t)}{R_A(t)}.
\end{aligned}
$$

$$f(t) = \varrho(t) R(t) = \alpha\beta t^{\beta-1} e^{-\alpha t^{\beta}}.$$



12.1. Theorem. Let X be a random variable with failure density f, reliability function R and hazard rate ϱ. Then

$$R(t) = e^{-\int_0^t \varrho(x)\, dx}.$$

A certain widget has a mean time between failures of 24 hours, i.e., failures occur at a constant rate of one failure every 24 hours.

One evening, the widget was observed to be working at 10 pm and then left unobserved for the night. The next morning at 6 am, it was observed to have failed earlier. What is the probability that it was still working at 5 am that morning?

Weibull Distribution

12.3. Definition. A random variable (X, f_X) is said to have a Weibull distribution with parameters α and β if its density is given by

$$f(x) = \begin{cases} \alpha\beta x^{\beta-1} e^{-\alpha x^{\beta}}, & x > 0, \\ 0, & \text{otherwise}, \end{cases} \qquad \alpha, \beta > 0.$$

$$\mu = \alpha^{-1/\beta}\Gamma(1 + 1/\beta)$$

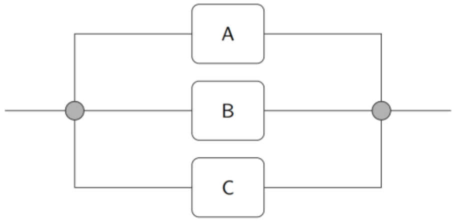
$$\sigma^2 = \alpha^{-2/\beta}\Gamma(1 + 2/\beta) - \mu^2.$$

The Uniform Distribution

$$f_X(x) = \begin{cases} 1 & 0 \le x \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

Reliability of Series and Parallel Systems

$$R_{\text{series}}(t) = P[\text{no component fails before } t] = \prod_{i=1}^{k} R_i(t),$$



$$\begin{aligned} R_{\text{parallel}}(t) &= 1 - P[\text{all components fail before } t] \\ &= 1 - \prod_{i=1}^{k}(1 - R_i(t)). \end{aligned}$$

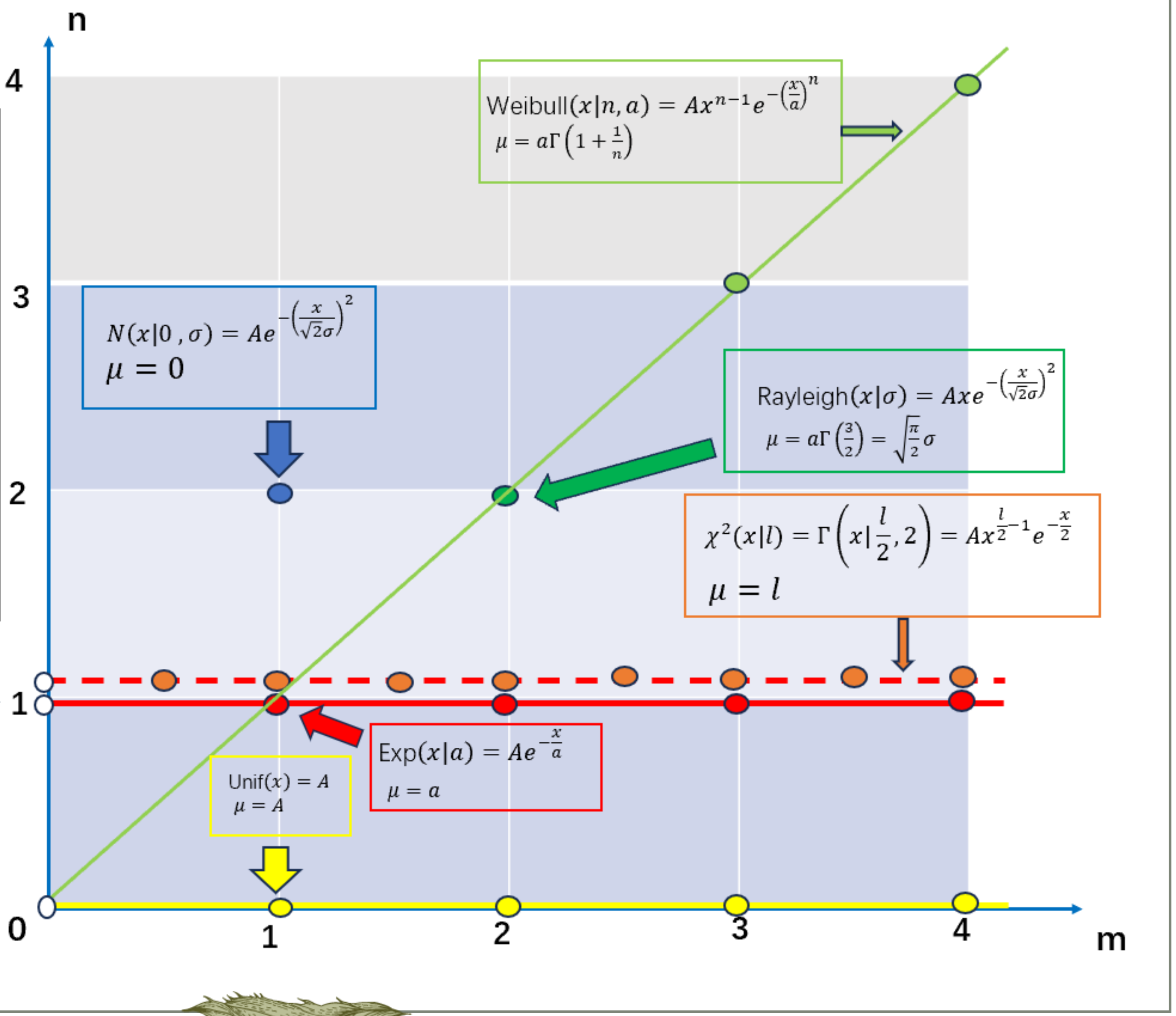
$$f(x) = Ax^{m-1}e^{-\left(\frac{x}{a}\right)^n}$$

8 Common Continuous Distributions

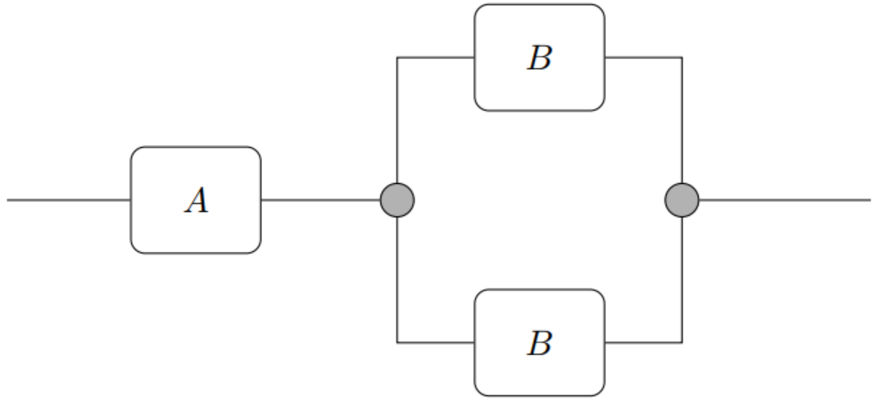
If the continuous distribution density function can be written in the form $f(x) = Ax^{m-1}e^{-\left(\frac{x}{a}\right)^n}$ (where A is a normalization constant), it can be positioned on an (m,n) coordinate system. The figure below shows the coordinate map of 8 common continuous distributions, with the constant coefficients of each distribution density function simplified and uniformly represented by "A" .
In the figure, μ represents the expectation (the mean) of the distribution.

$$\Gamma(x|m,a) = Ax^{m-1}e^{-\frac{x}{a}}$$
for $\forall a, \mu = ma$

$$\beta(x|m,1) = Ax^{m-1}$$
$$\mu = \frac{m}{m+1}$$



Consider the following system of components:



The system will fail if either component A or both components marked B fail. The components A and B have failure densities

$$f_A(t) = \frac{1}{100}e^{-t/100}, \qquad f_B(t) = \frac{1}{50}e^{-t/50}, \qquad t \geq 0,$$

respectively.

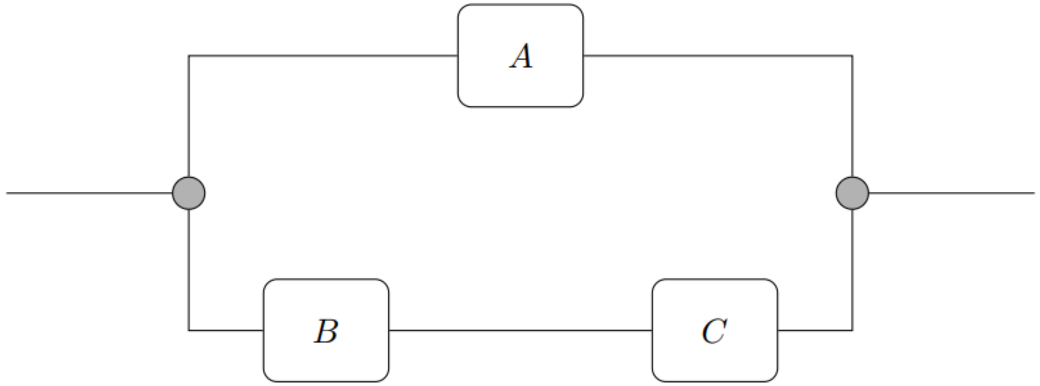
i) Find the reliability functions $R_A(t)$ and $R_B(t)$ of component A and each component B.

ii) Find the reliability function $R(t)$ of the system.

iii) What is the expected time of failure of each individual component?

iv) What is the expected time of failure of the system?

Consider the following system of components:



The components A, B and C have failure densities

$$f_A(t) = \frac{t}{50} e^{-t^2/100}, \qquad f_B(t) = \frac{1}{40} e^{-t/40}, \qquad f_C(t) = \frac{1}{25} e^{-t/25}, \qquad t \geq 0,$$

respectively.

i) Find the reliability functions $R_A(t)$, $R_B(t)$ and $R_C(t)$.

ii) Find the reliability function $R(t)$ of the system.

iii) What is the expected time of failure of each component? Evaluate all integrals explicitly; there should be no gamma functions or other integrals in your answer.

iv) What is the expected time of failure of the system? Evaluate all integrals explicitly; there should be no gamma functions or other integrals in your answer.

Suppose that X follows an exponential distribution. Then X^2 will follow

- ○ an exponential distribution $f_\beta(x) = \begin{cases} \beta e^{-\beta x}, & x > 0, \\ 0, & x \le 0, \end{cases}$
- ○ a chi-squared distribution $f_\gamma(x) = \begin{cases} \frac{1}{\Gamma(\gamma/2)2^{\alpha}} x^{\gamma/2-1} e^{-x/2}, & x > 0, \\ 0, & x \le 0, \end{cases}$
- ○ a Gamma distribution, but not a chi-squared distribution $f_{\alpha,\beta}(x) = \begin{cases} \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, & x > 0, \\ 0, & x \le 0, \end{cases}$
- ○ a Weibull distribution $f(x) = \begin{cases} \alpha\beta x^{\beta-1} e^{-\alpha x^\beta}, & x > 0, \\ 0, & \text{otherwise}, \end{cases} \quad \alpha, \beta > 0.$

Reference

- VE401 Slide, Horst
- Sheng, Z., Xie, S., & Pan, C. (2018). Probability Theory and Mathematical Statistics (5th ed.). Zhejiang University Press.
- VE401 SU25 Midterm Exam